

1. O/N/23/21/Q3

When the temperature of a liquid increases, the kinetic energy of its particles increases and the liquid expands.

- (a)** Explain, by referring to particles, why a liquid expands when heated.

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..... [2]

- (b)** The heater in a kettle is near to the base of the kettle. The kettle is filled with water at a temperature of 17°C and the heater is switched on.

- (i)** State the name of the method of thermal energy transfer that causes energy to be transferred to the water that is touching the heater.

..... [1]

- (ii)** Explain how thermal energy is transferred through all of the water in the kettle.

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.....

.....

..... [3]

- (iii)** State the boiling temperature of water at standard atmospheric pressure. Include the unit in your answer.

..... [1]

- (iv)** The kettle contains 2.5 kg of water at standard atmospheric pressure.

The specific heat capacity of water is $4200\text{ J/(kg }^{\circ}\text{C)}$.

When the water reaches its boiling temperature, the kettle switches off.

Calculate the increase in the internal energy of the water.

increase in internal energy = J [3]

2. O/N/22/22/Q3

Fig. 3.1 shows a man standing underneath an outdoor heater on a cold evening.

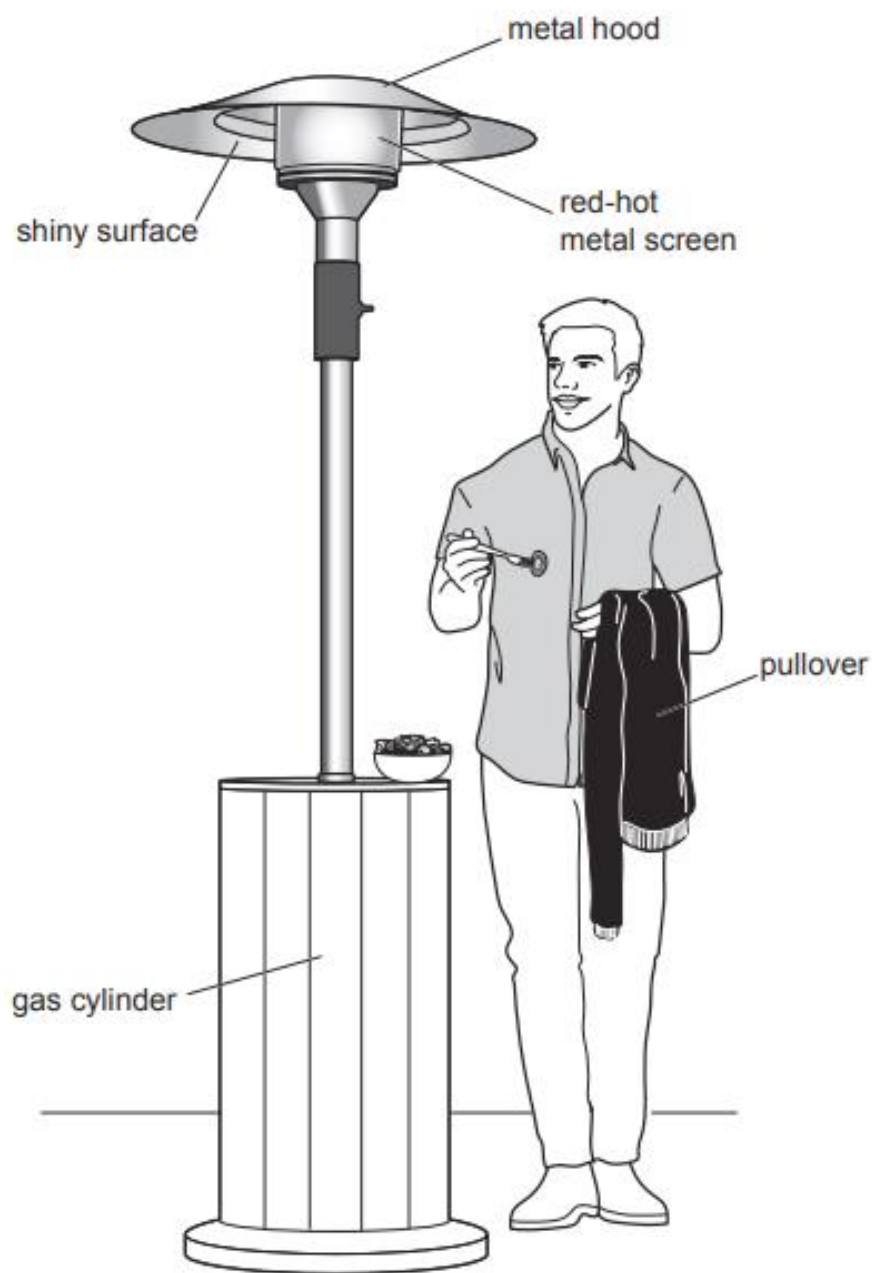


Fig. 3.1

Gas in the cylinder at the base of the heater is the fuel for the heater. When the heater is operating, the gas travels to the top of the heater where it burns.

(a) State the form of energy stored in the gas that is transferred by the heater.

..... [1]

(b) A metal screen surrounding the burning gas is heated by the burning gas until it is red-hot. The hot metal screen warms the man who is standing underneath it.

(i) Describe how thermal energy in the red-hot metal screen is transferred to the man and how it warms him.

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..... [3]

(ii) At the top of the heater is a metal hood that has a shiny lower surface.

Explain why this makes the energy transfer from the metal screen more efficient.

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..... [2]

(iii) The air temperature decreases and the man puts on a black pullover.

Explain **one** way in which wearing the black pullover helps to keep the man warm.

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..... [2]

[Total: 8]

3. M/J/21/21/Q2

Fig. 2.1 shows a hollow metal cube filled with boiling water. The temperature of the four vertical surfaces are equal but each surface has a different colour or texture.

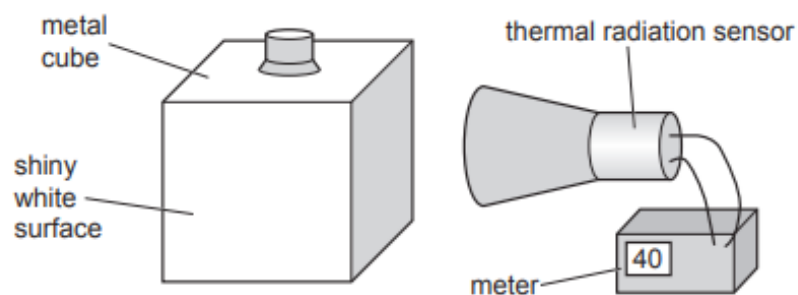


Fig. 2.1

A thermal radiation sensor is placed the same distance from each surface and the meter reading measures the thermal radiation emitted from each surface.

(a) The results are shown in **Fig. 2.2**.

Draw a line linking each type of surface with the appropriate meter reading. One line has been drawn for you.

type of surface		meter reading
dull black		40
dull white		60
shiny black	—	80
shiny white		100

Fig. 2.2

[2]

(b) The hot surfaces produce a convection current in the air outside the metal cube.

Describe how the convection current is produced.

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..... [3]

4.O/N/20/22/Q3

Fig. 3.1 shows a hot water tank that contains two electric heaters X and Y.

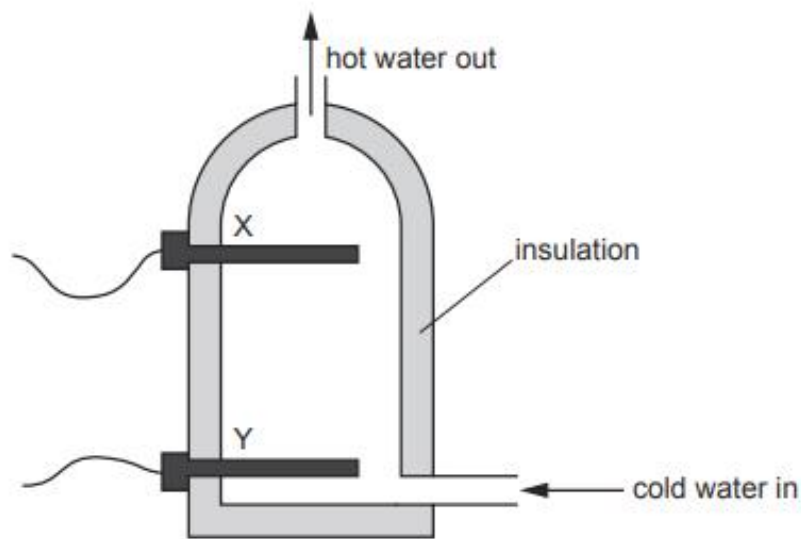


Fig. 3.1

Heater X is used during the daytime but heater Y is only used at night when electricity is cheaper.

- (a) The tank is full of cold water and X is switched on.

The temperature of **all** the water above X increases very quickly but the temperature of the water below X increases much more slowly.

- (i) Explain the process that causes the water above X to increase in temperature.

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..... [3]

- (ii) Heater Y remains switched off. Explain why the temperature of the water below heater X increases much more slowly than the temperature of the water above heater X.

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..... [2]

- (b)** The hot water tank is covered in a thick layer of insulating material. The material is a plastic that contains a large number of small pockets of trapped air.

Explain why this material is a good insulator.

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..... [2]

[Total: 7]

5. M/J/20/22/Q5(b)

Fig. 5.1 shows a microwave oven used to heat soup. The container for the soup is a glass bowl.

Microwaves created inside the oven are reflected by the metal walls.



Fig. 5.1

- (i) Explain why the choice of material for the container is important in microwave cooking.

.....
.....
..... [1]

- (ii) The soup is mostly water.

Microwaves are completely absorbed by a few centimetres of water.

As a result, microwaves do not reach the centre of the soup.

The instructions suggest that, after the microwave oven is turned off, the soup:

- is not stirred
- is left for some minutes so that the centre becomes hot.

State the name of and describe each of the **two** processes by which thermal energy transfers throughout the soup after the microwave oven is turned off.

name of first process

description

.....

.....

name of second process

description

.....

..... [3]

6. M/J/20/21/Q4

Glass and iron are both conductors of heat. However, glass is a poor conductor of heat and iron is a good conductor of heat.

- (a) Describe, using ideas about particles, how the conduction of heat takes place in glass and in iron. You should make it clear why iron is a better conductor of heat.

conduction in glass

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conduction in iron

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[4]

(b) Fig. 4.1 shows apparatus used to show expansion.

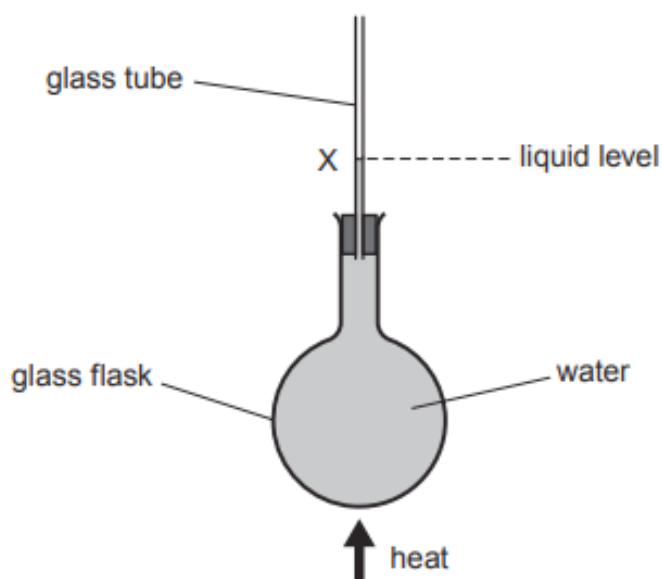


Fig. 4.1

The glass flask, full of water, is heated. A student is surprised when the liquid level X in the glass tube falls for a few seconds before it rises.

(i) Suggest why the liquid level falls and why it then rises.

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..... [2]

(ii) Describe how heat is transferred throughout the water in the glass flask.

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..... [2]

7. O/N/19/21/Q9

Fig. 9.1 shows the structure of a water cooler that is used to supply cold water to the workers in a hot office.

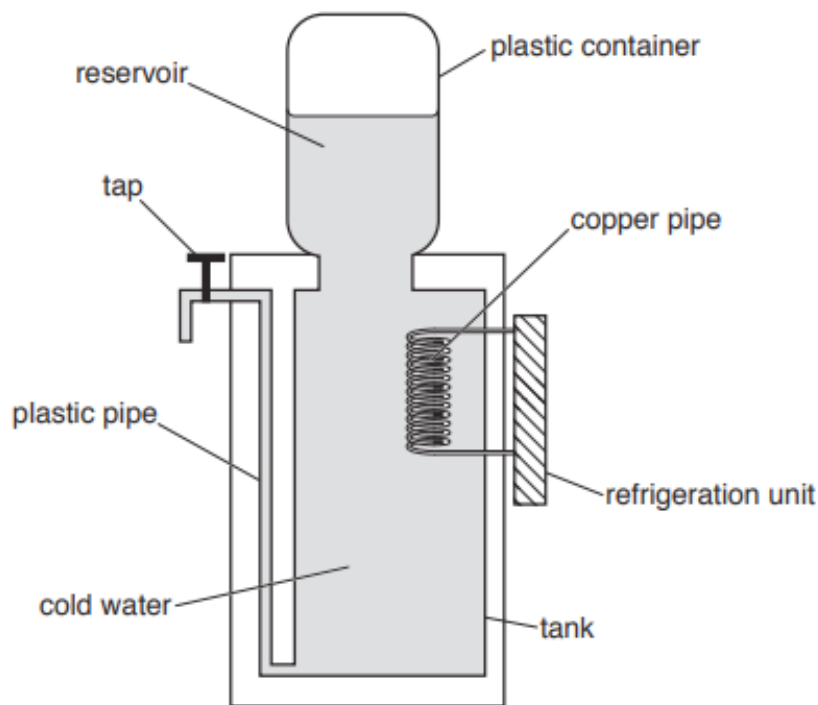


Fig 9.1

- (a) The reservoir of water in the plastic container is at room temperature. It does not mix with the cold water in the tank below.

Explain why.

.....
.....
..... [2]

- (b) When the tap is opened, water at room temperature from the reservoir flows down into the tank. Cold water from the tank flows through the plastic pipe and out of the tap.

Cold liquid from the refrigeration unit is pumped through the copper pipe and thermal energy passes through the copper to this liquid.

- (i) Suggest why this pipe is made from copper.

.....
..... [1]

- (ii) Explain, in terms of free electrons, how thermal energy is transferred through the copper.

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..... [3]

- (iii) As the water near the copper pipe cools, it begins to mix with the rest of the water in the tank.

Explain how this happens.

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..... [3]

- (c) The tap is opened and water at 21°C flows from the reservoir into the tank.

The specific heat capacity of water is $4200\text{ J}/(\text{kg}^{\circ}\text{C})$.

Calculate the thermal energy that is removed from 250 g of this water to reduce its temperature to 7°C .

energy = [3]

- (d) (i) Describe the motion of the molecules in a liquid which is at a uniform temperature throughout.

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..... [2]

- (ii) State what happens to the motion of the molecules in a liquid as the temperature of the liquid decreases.

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..... [1]

8. M/J/19/22/Q10

Fig. 10.1 is a diagram of a soldering iron. Solder is a mixture of metals used to make a permanent contact between electrical wires.

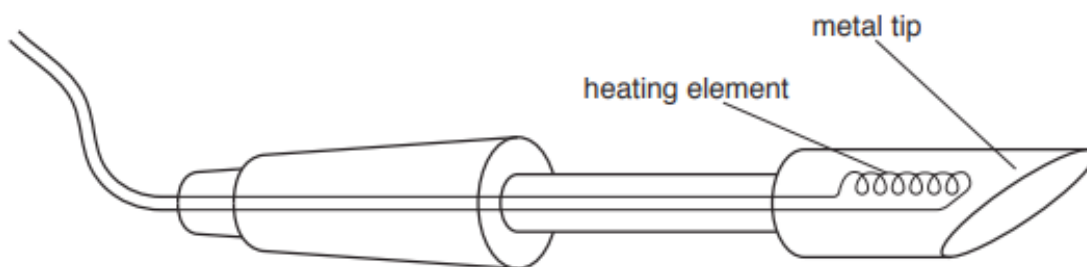


Fig. 10.1

The heating element raises the temperature of the metal tip. When solder is placed against the tip, the solder melts over the wires to be joined. When the solder cools, it solidifies and the permanent connection is made.

The working temperature of the metal tip is 380°C .

(a) The boxes in Fig. 10.2 show two materials and some different melting points.

material	melting point
metal of the metal tip	1000°C
	380°C
	200°C
solder	20°C
	0°C

Fig. 10.2

On Fig. 10.2, draw a line from the metal of the metal tip and a line from the solder to a suitable melting point for each. [2]

- (b) The heating element is rated at 24 V, 3.3A.

The heating element is switched on.

The temperature of the metal tip rises from 20 °C to 320 °C in the first 10 s.

- (i) Calculate the electrical energy supplied to the heating element in the first 10 s.

energy = [2]

- (ii) The metal tip is made of copper and has a mass of 2.3g. The specific heat capacity of copper is 0.39 J/(g °C).

Calculate the thermal energy (heat) gained by the metal tip in the first 10 s.

thermal energy = [3]

- (c) (i) Describe, in terms of free electrons, the process by which heat transfers through the metal tip.

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.....
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..... [2]

- (ii) Heat is lost from the metal tip by convection in the air.

Explain how convection occurs in the air.

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..... [2]

9. O/N/18/22/Q3

Fig. 3.1 shows a cylindrical copper kettle that contains cold water.

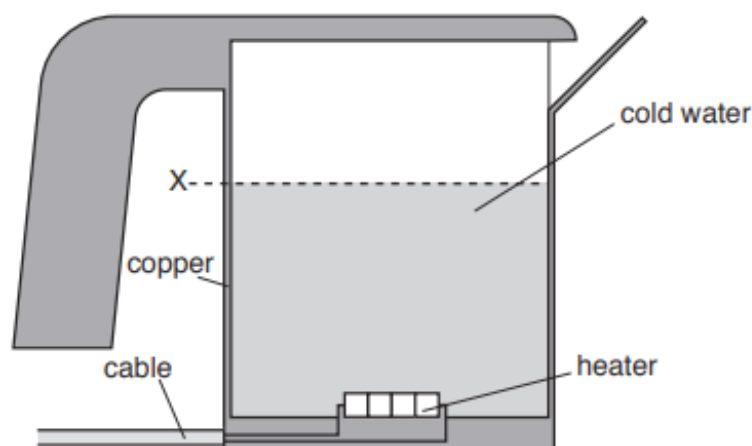


Fig. 3.1

The kettle is used to heat water and there is an electric heater at the base.

(a) State and explain the advantage of heating the water from below.

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..... [2]

(b) As the water is heated, it expands.

(i) Explain, in terms of molecules, why water expands when it is heated.

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.....

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..... [2]

(ii) Copper also expands when heated.

State what happens to level X of the water in the kettle. Explain your answer in terms of the expansion of the copper and the water.

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..... [1]

10. O/N/18/22/Q4

A house has several solar panels on the roof.

These panels use energy from the Sun both to generate electricity and to raise the temperature of water that passes through tubes inside the panels.

(a) The panels on the roof of the house have a black surface.

(i) State how energy from the Sun travels through space before it reaches the Earth.

.....
..... [1]

(ii) Explain the advantage of using panels that have a black surface.

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.....
.....
..... [2]

(b) On one occasion, the panels are supplying an electric current of 15A at a voltage of 24 V.

(i) Calculate the electrical energy generated by the panels in one hour.

electrical energy = [2]

(ii) In the same time, 51 kg of cold water is pumped through the panels. The temperature of the water increases from 16°C to 45°C.

The specific heat capacity of water is 4200 J/(kg °C).

Calculate the increase in thermal energy of the water.

thermal energy increase = [3]

11. M/J/17/21/Q3

Fig. 3.1 shows a metal coffee cup on a metal warming plate.

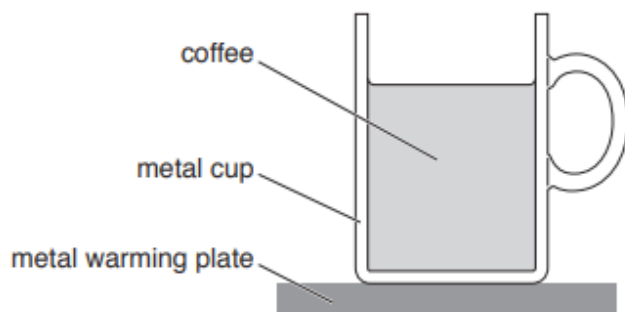


Fig. 3.1

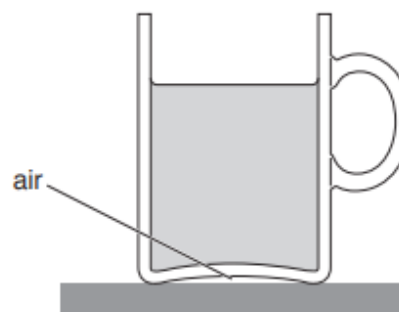


Fig. 3.2

There is a small electrical heater inside the warming plate that keeps the plate hotter than the coffee.

- (a) Describe how heat is transferred through the metal and then to all of the liquid in the cup.

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.....[3]

- (b) A cup of a different shape is placed on the same heater, as shown in Fig. 3.2. The two cups are made of the same metal and contain the same amount of coffee.

Explain why the coffee in the cup in Fig. 3.2 is not kept as warm as the coffee in the cup in Fig. 3.1.

.....

.....[1]

- (c) The outside surface of the cup can be either black or white and can be either dull or shiny.

- (i) Underline which colour **and** which type of surface is best to keep the coffee warm.

black **white** **dull** **shiny** [1]

- (ii) Explain your answer to (c)(i).

.....

.....

.....[1]

12. O/N/15/21/Q2

Fig. 2.1 shows a freezer for keeping food frozen.



Fig. 2.1

- (a) The outside surface of a freezer is painted white.

Explain one advantage of having the outside surface of the freezer painted white.

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.....

..... [2]

- (b) (i) The lid of the freezer is closed and air at room temperature is trapped inside the freezer. The freezer is switched on.

State and explain, using ideas about molecules, what happens to the pressure of the air in the freezer, as it cools.

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.....

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..... [3]

- (ii) When the freezer reaches its operating temperature, it is more difficult to open the lid.

Explain why.

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.....

..... [1]

13. O/N/15/22/Q4

A copper saucepan with a wooden handle contains cold water. The saucepan is placed on a red-hot heating element that is a part of an electric cooker.

Fig. 4.1 shows the saucepan on the heating element.

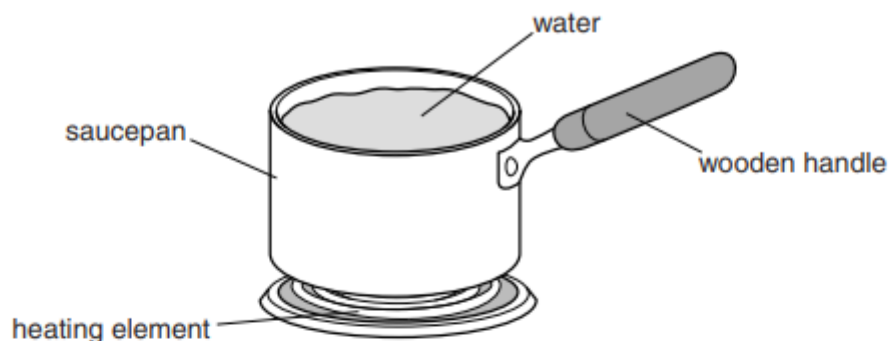


Fig. 4.1

- (a) Explain why wood is a suitable material for the handle of the saucepan.

..... [1]

- (b) (i) Describe and explain, in terms of free electrons, how thermal energy is transferred through the copper base of the saucepan.

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.....
.....
.....
..... [3]

- (ii) Describe and explain how thermal energy is transferred upwards through the water.

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.....
..... [3]

14. O/N/14/21/Q10

The casing of an electric kettle is made of white plastic. Fig. 10.1 shows the heating element positioned in the base of the kettle.

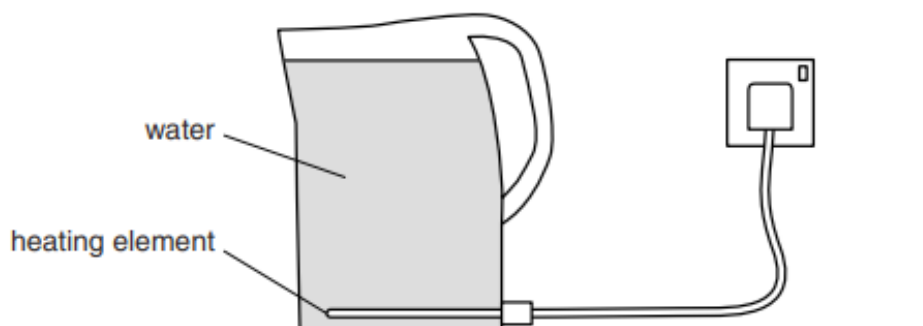


Fig. 10.1

- (a) (i) The heating element supplies thermal energy to the water at the bottom of the kettle.

Describe and explain how the thermal energy is transferred throughout the water.

.....

[3]

- (ii) Explain why a kettle with its heating element in the water at the top of the kettle does not heat the water uniformly.

.....
[1]

- (b) The kettle is powered by a 230V supply. It is switched on for 3.5 minutes and there is a current of 9.6 A in the heating element.

- (i) Calculate the thermal energy produced in the heating element in this time.

thermal energy =[2]

- (ii) The kettle contains 1.6 kg of water that was at an initial temperature of 22°C. The specific heat capacity of water is 4200 J/(kg °C).

Calculate the maximum possible temperature of the water.

temperature =[3]

- (iii) Suggest one reason why the temperature of the water, after 3.5 minutes, is less than the value calculated in (b)(ii).

.....
.....[1]

- (c) Explain one advantage of

- (i) using plastic for the casing of a kettle,

.....
.....[1]

- (ii) choosing white as the colour for the outside of the casing.

.....
.....[1]

- (d) The kettle is switched on again and the water reaches its boiling point. It starts to boil and the kettle remains switched on.

- (i) State the meaning of *boiling point*.

.....
.....[1]

- (ii) Explain, in terms of molecules, what happens to the thermal energy that is supplied when the water is boiling.

.....
.....
.....[2]

15. M/J/14/22/Q10

Two metal saucepans contain the same mass of hot water at the same initial temperature. Pan A is white and pan B is black, but otherwise the two saucepans are identical. Both saucepans are uncovered and cool under the same conditions. The cooling curves for the two saucepans are shown in Fig. 10.1.

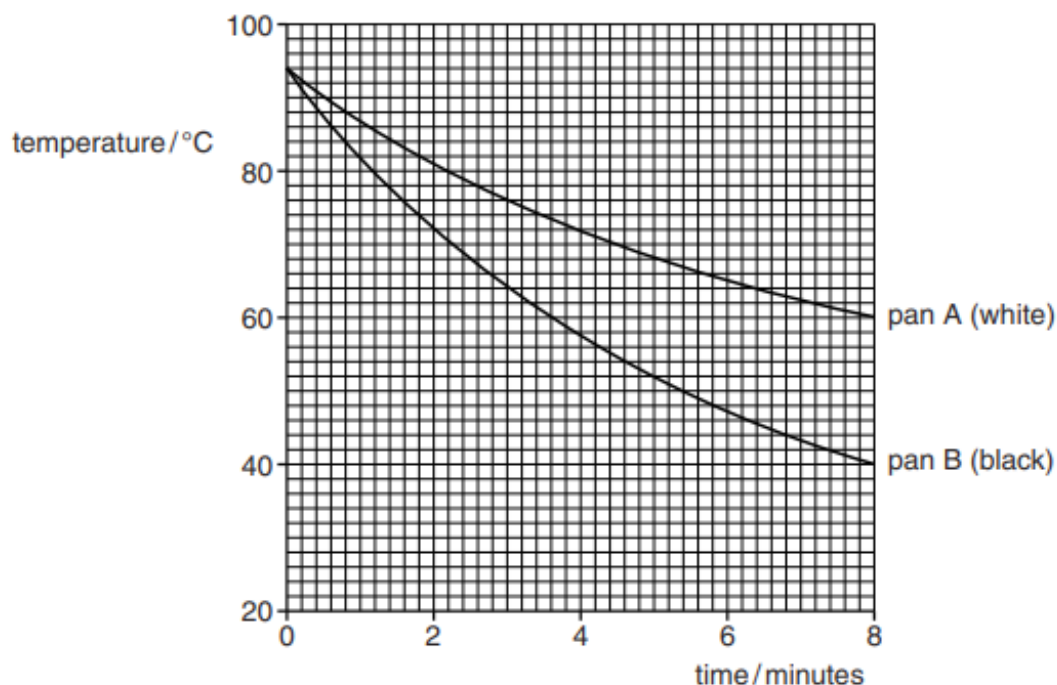


Fig. 10.1

(a) Describe how the water in a pan loses heat by

(i) conduction,

.....
.....
..... [2]

(ii) convection.

.....
.....
..... [2]

(b) (i) Explain why pan B cools faster than pan A.

.....
..... [1]

- (ii) Describe and explain how Fig. 10.1 is different when the pans are covered and the experiment is repeated.

.....
.....
..... [2]

- (c) The specific heat capacity of water is $4200 \text{ J/(kg } ^\circ\text{C)}$.

- (i) Explain what is meant by *specific heat capacity*.

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.....
..... [2]

- (ii) The specific heat capacity of water is very high. Suggest one disadvantage of this when water is used for cooking.

.....
..... [1]

- (iii) The water in pan A cools for 8 minutes, as shown in Fig. 10.1. During this time, the water loses an average of 9000 J of thermal energy per minute.

1. Calculate the mass of water in pan A.

mass = [3]

2. The mass of water in pan B is the same as that in pan A.

Calculate the thermal energy lost from the water in pan B during the 8 minutes.

loss of thermal energy = [2]

16. O/N/13/21/Q4

Fig. 4.1 shows food being cooked in an electric grill.

red-hot heating elements

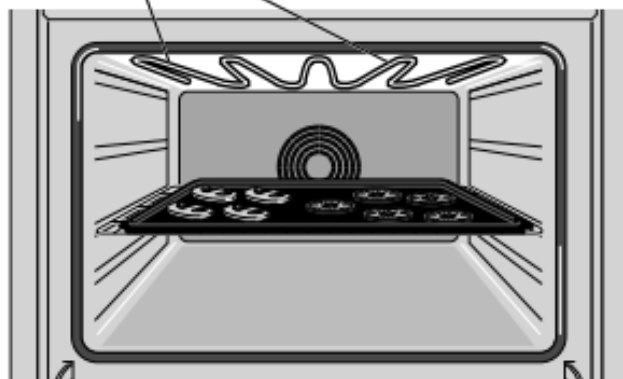


Fig. 4.1

There are red-hot heating elements above the food and thermal energy (heat) is transmitted to the food by radiation.

(a) Explain what is meant, in this case, by *radiation*.

.....
.....
.....[2]

(b) Explain why very little thermal energy is transmitted to the food by

(i) conduction,

.....
.....[1]

(ii) convection.

.....
.....[1]

17. O/N/12/21/Q10

Fig. 10.1 shows a domestic heater that is used to heat a room. It contains 16 large bricks each of mass 7.5 kg.

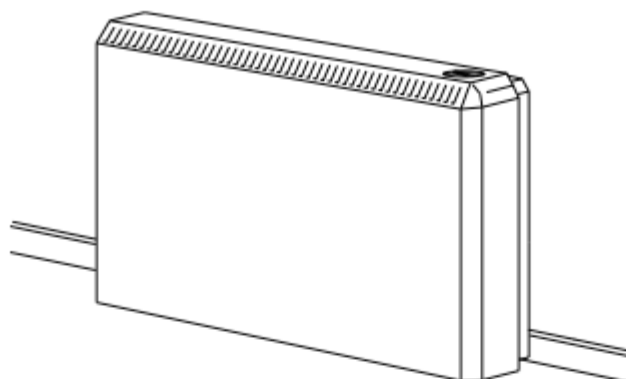


Fig. 10.1

During the night, the bricks are heated from a temperature of 17°C to 96°C . The bricks are made from a material that has a specific heat capacity of $2300\text{ J}/(\text{kg }^{\circ}\text{C})$.

- (a) Calculate the thermal energy (heat) supplied to the bricks.

thermal energy = [3]

- (b) During the day, the bricks gradually cool and the stored thermal energy is released to the room. After 7.0 hours, the bricks have cooled to 17°C .

- (i) Calculate the average rate of release of thermal energy to the room during these 7.0 hours.

rate = [2]

- (ii) At the beginning of the day, the heater releases thermal energy at a greater rate than later in the day.

Suggest why.

.....
.....
..... [2]

- (c) Fig. 10.2 is a diagram of the heater standing on the floor of the room and placed next to one of the walls.



Fig. 10.2

Describe, and explain in detail, how the thermal energy is transferred throughout the room. You may draw on Fig. 10.2.

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..... [5]

- (d) State one method of thermal insulation that is used to keep a room warm, and explain why it is effective.

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..... [3]

18. O/N/12/22/Q5

- (a)** No thermal energy (heat) is transferred from the surface of the Sun to the Earth by either conduction or convection.

Explain why this is so.

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.....

.....[2]

- (b)** In a certain country, the climate is very sunny and hot during the day and extremely cold during the night.

Explain how painting the houses white helps to maintain a comfortable temperature both during the day and during the night.

during the day:

.....

.....

during the night:

.....

.....

[3]

19. M/J/12/22/Q3

Fig. 3.1 shows the plan of a bedroom and part of the main room of a house. Other rooms are not shown.

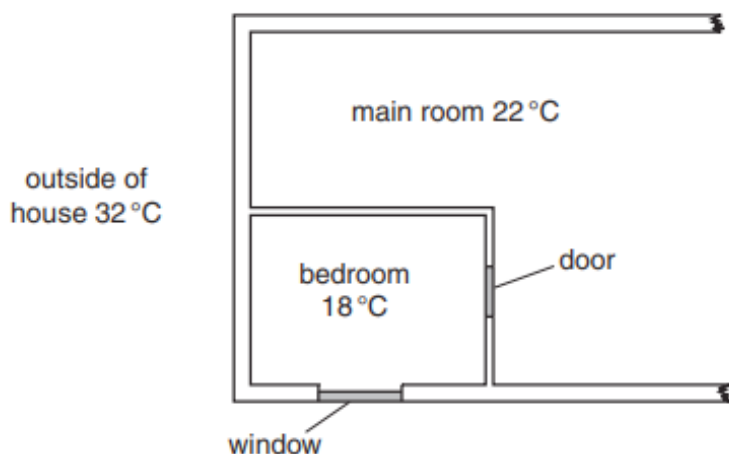


Fig. 3.1

The temperatures of the main room, the bedroom and the outside of the house are shown on Fig. 3.1.

Fig. 3.2 shows all the thermal energy (heat) inputs to the bedroom in one hour.

thermal energy input to bedroom	
through door and walls from main room	50 000 J
through walls from outside of house	2 000 000 J
through window	1 000 000 J
from person sleeping in bedroom	250 000 J

Fig. 3.2

- (a) Suggest why more thermal energy enters the bedroom from the outside of the house than from the main room.

.....[1]

- (b) An air conditioner keeps the temperature constant in the bedroom by removing thermal energy.

- (i) Calculate the total thermal energy that the air conditioner removes from the bedroom in one hour.

thermal energy =[1]

- (ii) The electrical power input to the air conditioner is 300W.
Calculate the electrical energy input into the air conditioner in 1 hour.

energy =[2]

- (c) The air conditioner cools the air at the top of the room. This causes a convection current in the room.

Explain how the cold air gives rise to the convection current.

.....
.....
.....[3]

20. M/J/10/22/Q3

Fig. 3.1 shows a metal roof. One side is facing the Sun.

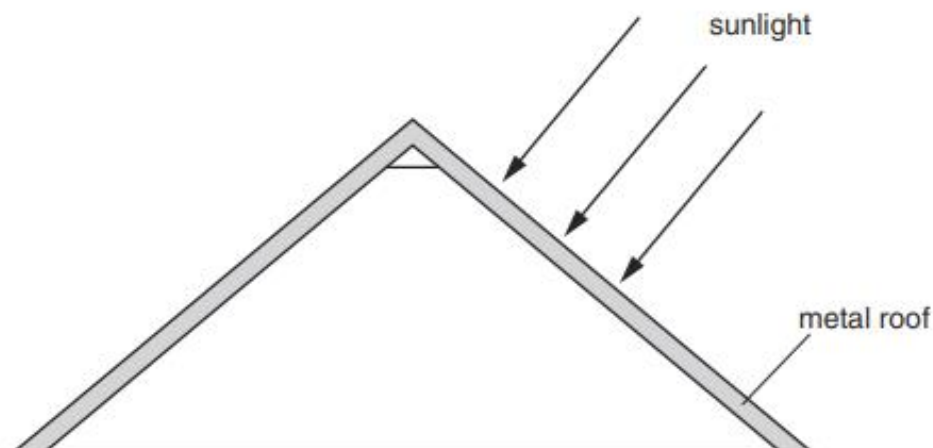


Fig. 3.1

- (a) State the means by which thermal energy (heat) is transferred from the Sun to the Earth and explain why other means of thermal energy transfer are not involved.

.....
..... [2]

- (b) Describe how thermal energy is transferred through the metal roof from the heated surface.

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.....
..... [2]

- (c) During the night, the metal roof loses $1.2 \times 10^6 \text{ J}$ of thermal energy and its temperature falls by 20°C .

The specific heat capacity of the metal in the roof is $400 \text{ J}/(\text{kg}^\circ\text{C})$.

Calculate the mass of metal in the roof.

mass = [2]

21. M/J/09/Q2

Fig. 2.1 shows a metal pan containing water on a cooker. The hotplate heats the water.

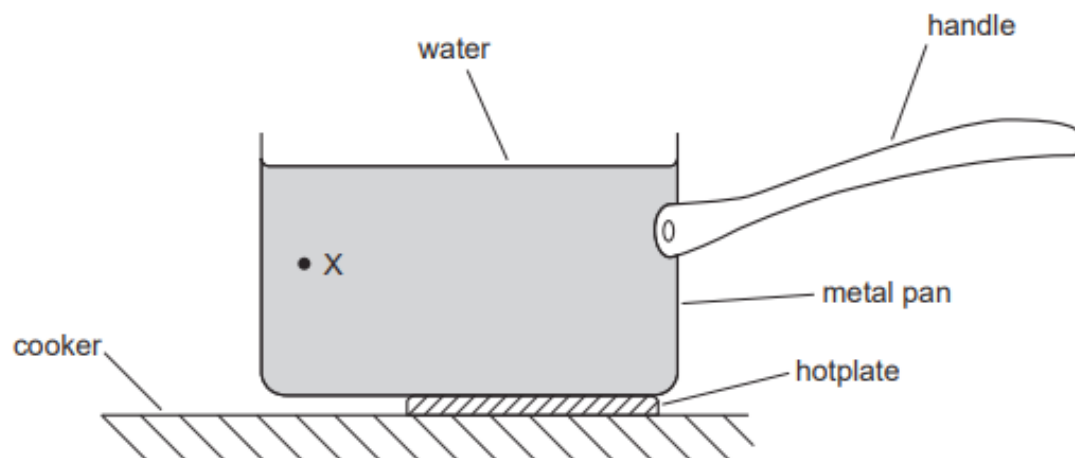


Fig. 2.1

- (a) (i) State the method of heat transfer through the metal pan.

..... [1]

- (ii) Describe how the molecules transfer heat through the metal pan.

.....
.....
..... [1]

- (b) (i) On Fig. 2.1, draw an arrow to show the direction of movement of the water at point X. [1]

- (ii) Explain why the water moves in this direction.

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.....
.....
..... [3]

22. M/J/08/Q3

During a marathon race, the runner shown in Fig. 3.1 is very hot.



Fig. 3.1

At the end of the race, evaporation and convection cool the runner.

- (a) (i)** Explain how evaporation helps the runner to lose energy. Use ideas about molecules in your answer.

.....

.....

.....

..... [2]

- (ii)** Explain why hot air rises around the runner at the end of the race.

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..... [1]

- (b)** At the end of the race, the runner is given a shiny foil blanket, as shown in Fig. 3.2. Wearing the blanket stops the runner from cooling too quickly.



Fig. 3.2

Explain how the shiny foil blanket helps to reduce energy losses.

Use ideas about conduction, convection and radiation in your answer.

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..... [3]

23. O/N/07/Q4

Fig. 4.1 shows equipment placed on top of a house that uses solar energy to produce hot water.

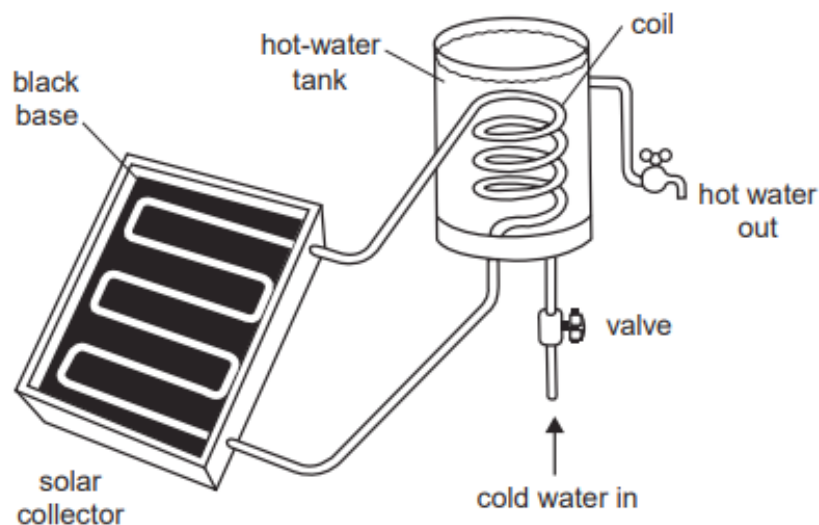


Fig. 4.1

(a) Explain why the solar collector has a black base.

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..... [2]

(b) State and explain why the hot water in the solar collector travels to the hot-water tank.

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..... [2]

(c) Fig. 4.1 does not show any insulation.

(i) Explain why it is important to insulate the hot-water tank.

.....

..... [1]

(ii) Explain how the hot-water tank is insulated.

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..... [1]

24. O/N/06/Q3

Three horizontal rods are placed with one end just above a Bunsen flame. The other end of each rod is coated with wax, as shown in Fig. 3.1.

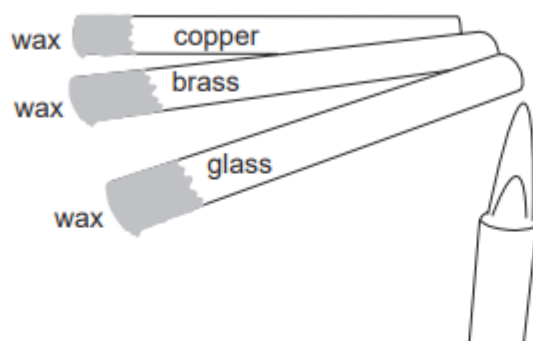


Fig. 3.1

- (a) Describe how you would use the apparatus to discover which rod is the best conductor of heat.

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..... [2]

- (b) Two metal teapots are identical except that one is black on the outside and the other is white on the outside, as shown in Fig. 3.2.



Fig. 3.2

The teapots each contain the same amount of hot water.

State and explain which teapot will cool down more quickly.

.....

.....

.....

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..... [3]

25. M/J/04/Q2

Heat is transferred by conduction, convection and radiation.

- (a) (i) State which of the three methods is responsible for the transfer of heat from the Sun to the Earth.

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- (ii) Explain why the other two methods cannot be involved in this transfer.

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[2]

- (b) A hand feels hot when placed above a lighted match, as shown in Fig. 2.1.
Explain in detail how convection causes this to happen.

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Fig. 2.1 [2]

- (c) Fig. 2.2 shows a layer of fibreglass placed between the ceiling of a room and the roof of a house.

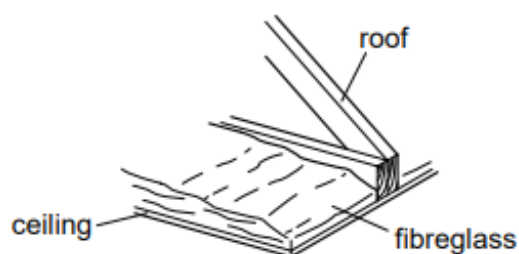


Fig. 2.2

Explain how the layer of fibreglass helps to keep the room warm when it is cold outside.

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.....[2]